# Handling Imbalanced Data in Binary Classification: A Review and Comparative Experimental Study

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## Abstract

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Imbalanced Data, Resampling Methods, Boosting Algorithms, Machine Learning

## 1 Introduction

### 1.1 Background and Problem Definition

In recent years, the rapid expansion of data availability across large-scale, complex, and networked systems has propelled machine learning to prominent roles in numerous application domains [1].

However, real-world data distributions are inherently complex and frequently exhibit a significant class imbalance. The class imbalance problem arises when the number of instances belonging to one class (the majority class) substantially outnumbers those of another (the minority class), resulting in a severely skewed training distribution [1, 2].

This phenomenon is pervasive in many high-stakes domains, including financial fraud detection, network intrusion detection, and the diagnosis of rare diseases such as diabetes and stroke, where the minority class—despite its low frequency of occurrence—typically constitutes the primary target of predictive interest [3, 4].

### 1.2 Challenges and the Pitfalls of Evaluation Metrics

Standard machine learning models are generally designed under the implicit assumption of an approximately balanced class distribution. Consequently, when applied to skewed data, these models exhibit a systematic bias toward the majority class, yielding poor predictive performance for minority instances—precisely those of greatest practical concern [1].

This issue is further compounded by the widespread use of conventional evaluation metrics. Overall accuracy, in particular, is highly misleading under class imbalance: a naive classifier that assigns all instances to the majority class can attain superficially high accuracy while entirely failing to identify any minority-class example [5].

To overcome this limitation, the research community has advocated the adoption of more discriminative metrics, including the F1-Score, the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) [6], and the Geometric Mean (G-Mean) [7], which collectively provide a more faithful assessment of model performance across both classes.

### 1.3 Taxonomy of Imbalanced Learning Techniques

To address the challenges posed by imbalanced datasets, a rich body of literature has proposed numerous techniques, which are broadly categorized into three main families: data-level methods, algorithmic-level methods, and hybrid strategies [1, 8].

Data-level approaches modify the training data distribution prior to model induction, either by oversampling the minority class—most notably through the Synthetic Minority Over-sampling Technique (SMOTE) [2] and its variants—or by undersampling the majority class through methods such as Random Under-Sampling (RUS).

Algorithmic-level approaches, by contrast, leave the data distribution unchanged and instead modify the learning procedure itself; representative techniques include cost-sensitive learning and ensemble-based methods such as weighted classifiers and boosting frameworks [9].

Finally, hybrid strategies combine the complementary strengths of both data resampling and algorithmic robustness, with methods such as SMOTEBoost and RUSBoost serving as prominent examples [9].

### 1.4 Research Gap and Objectives

Despite the substantial volume of existing work, many comparative studies evaluate imbalanced learning techniques across heterogeneous datasets that vary simultaneously in multiple dimensions—including class overlap, feature complexity, and sample size—making it difficult to isolate the independent effect of the Imbalance Ratio (IR) on classifier performance [1]. Confounding factors of this nature limit the interpretability of empirical findings and reduce their practical utility as guidelines for practitioners.

To address this gap, this study proposes a rigorous empirical framework grounded in controlled experimental design. Rather than relying solely on a diverse collection of independent datasets, we systematically vary the IR within a curated set of benchmark datasets spanning a wide range of imbalance severity—from mild ($IR \approx 2.4$) to extreme ($IR \approx 41.4$)—thereby enabling direct, controlled comparisons across escalating levels of class imbalance.

Within this framework, we conduct a systematic evaluation of representative methods from all three technique families—data-level, algorithmic-level, and hybrid—using three established base classifiers (SVM, Random Forest, and XGBoost) and a comprehensive set of evaluation metrics (F1-Score, AUC-ROC, and G-Mean). By analyzing the performance degradation trajectories of each strategy as class imbalance worsens, this study aims to provide both a nuanced empirical understanding of the strengths and limitations of existing approaches and actionable practical guidance for selecting appropriate imbalanced learning techniques in binary classification tasks.

# 2 Method

## 2.1 Data-level Methods

At the data pre-processing stage, techniques for addressing class imbalance can be broadly categorized into two types: non-synthetic and synthetic methods. Non-synthetic methods adjust the class distribution using existing data, leaving the underlying information unaltered. Conversely, synthetic methods mitigate the imbalance by generating new data samples using existing features.

### 2.1.1 *Non-synthetic method.*

*(i) Random sampling:* Resampling strategies generally include techniques such as under-sampling, over-sampling, and hybrid sampling. Random under-sampling (RUS) deletes samples from the majority class, which may discard useful data. Random over-sampling (ROS) achieves dataset balance by duplicating minority class samples, but this exact duplication increases the risk of overfitting. [81, 10]

In most cases, under-sampling improves model performance more effectively than over-sampling. However, in certain specific scenarios—such as when the negative class is dominant—the opposite holds true. The optimal resampling ratio varies depending on the dataset and the chosen resampling strategy, rather than simply supplementing or reducing classes to a perfectly balanced state. [44, 11]

*(ii) Hybrid sampling:* Hybrid sampling combines the advantages of both over-sampling and under-sampling while mitigating their respective disadvantages. Applying over-sampling before under-sampling generally yields better performance than the reverse order. However, the hybrid sampling ratio must be determined manually, often requiring domain knowledge. If this ratio is inappropriately set for a given dataset, the approach may fail to show the benefits of hybridization and could even perform worse than using either technique alone. [12]

# 3 EXPERIMENTAL SETUP

## 3.1 Datasets

To rigorously investigate the effect of varying Imbalance Ratios (IR) on classifier performance, this study employs eight benchmark datasets sourced from the KEEL (Knowledge Extraction based on Evolutionary Learning) repository [**?** ], a widely adopted resource for imbalanced classification research. The selected datasets span a broad spectrum of imbalance severity—from mild to extreme—and originate from diverse real-world domains, thereby enabling controlled cross-IR comparisons while maintaining ecological validity.

The Imbalance Ratio is defined formally as:

$$\mathrm{IR} = \frac{|S_{\mathrm{maj}}|}{|S_{\mathrm{min}}|}, \tag{1}$$

where $|S_{\mathrm{maj}}|$ and $|S_{\mathrm{min}}|$ denote the cardinalities of the majority and minority classes, respectively. Following the severity categorization proposed in the imbalanced learning literature [**?** ], the datasets are stratified into four levels: *mild* ($\mathrm{IR} < 3$), *moderate* ($3 \leq \mathrm{IR} < 15$), *severe* ($15 \leq \mathrm{IR} < 30$), and *extreme* ($\mathrm{IR} \geq 30$).

All experiments employ **stratified 5-fold cross-validation** to ensure that the class distribution is preserved in every fold. The complete characteristics of the benchmark suite are summarised in Table 1.

The inclusion of four Yeast variants (Yeast-1, -3, -4, and -6) sharing an identical feature space and sample size ($n = 1{,}484$) is a deliberate methodological choice: by holding feature complexity and data size constant while varying only the IR, these four datasets allow the direct isolation of class imbalance severity as the sole experimental variable, effectively controlling for confounds that commonly hinder cross-dataset comparisons [**?** ].

## 3.2 Evaluation Metrics

A central challenge in evaluating classifiers on imbalanced datasets is metric selection. As demonstrated by He and Garcia [**?** ] and mathematically formalised by Luque et al. [**?** ], overall *Accuracy* is a fundamentally flawed metric under class imbalance: a degenerate classifier that assigns every instance to the majority class can attain superficially high accuracy while entirely failing to detect any minority-class example. Accordingly, this study excludes Accuracy as a primary performance measure and instead adopts a suite of imbalance-aware metrics grounded in the binary confusion matrix, whose elements are defined as follows: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), where the *positive* class denotes the minority class throughout.

*Precision and Recall.* Precision measures the proportion of predicted positive instances that are truly positive; Recall (Sensitivity) measures the proportion of actual positive instances that are correctly identified. Both metrics are defined over the minority class and are expressed as [**?** ]:

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \tag{2}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN}. \tag{3}$$

*F1-Score.* The F1-Score is the harmonic mean of Precision and Recall, providing a single scalar that penalises severe imbalances between the two [**?** ]:

$$F_1 = 2 \cdot \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} = \frac{2\,TP}{2\,TP + FP + FN}. \tag{4}$$

Because the harmonic mean is strictly lower than the arithmetic mean whenever Precision and Recall diverge, $F_1$ constitutes a conservative and informative composite measure for imbalanced classification tasks.

*AUC-ROC..* The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is a threshold-independent metric that quantifies a classifier's ability to discriminate between classes across all possible decision thresholds [**?** ]. Let the False Positive Rate (FPR) and True Positive Rate (TPR) be defined as:

$$\mathrm{FPR} = \frac{FP}{FP + TN}, \qquad \mathrm{TPR} = \frac{TP}{TP + FN}. \tag{5}$$

The AUC-ROC is then computed as:

$$\mathrm{AUC}_{\mathrm{ROC}} = \int_0^1 \mathrm{TPR}\big(\mathrm{FPR}^{-1}(x)\big)\, dx. \tag{6}$$

A value of 0.5 corresponds to random classification, while 1.0 denotes a perfect classifier. As noted by Saito and Rehmsmeier [**?** ], however, ROC-based metrics can present an overly optimistic

**Table 1: Summary of Benchmark Datasets.**

| Dataset | Instances | Features | Minority | Majority | Imbalance Ratio | Severity |
|---|---|---|---|---|---|---|
| Yeast-1 | 1,484 | 8 | 429 | 1,055 | 2.46 | Mild |
| Yeast-3 | 1,484 | 8 | 163 | 1,321 | 8.10 | Moderate |
| Yeast-4 | 1,484 | 8 | 51 | 1,433 | 28.10 | Severe |
| Yeast-6 | 1,484 | 8 | 35 | 1,449 | 41.40 | Extreme |

picture under extreme class imbalance, because the large pool of true negatives in the majority class suppresses the FPR, thereby inflating the apparent curve.

*G-Mean.* The Geometric Mean (G-Mean), originally proposed by Kubat and Matwin [? ] for imbalanced learning, is the square root of the product of Sensitivity (Recall) and Specificity. It provides a balanced assessment of performance across both classes simultaneously: poor performance on *either* class leads to a sharp decline in the overall score. Specificity and G-Mean are defined as:

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{7}$$

$$\text{G-Mean} = \sqrt{\text{Recall} \times \text{Specificity}} = \sqrt{\frac{TP}{TP + FN} \cdot \frac{TN}{TN + FP}}. \tag{8}$$

*Metric Selection Rationale.* Although this study reports all metrics described above for completeness, the **primary evaluation metrics are F1-Score, AUC-ROC, and G-Mean**. These three are selected because they collectively capture complementary dimensions of classifier performance under imbalance: the F1-Score reflects the precision–recall trade-off for the minority class; the AUC-ROC captures threshold-independent ranking ability across all decision boundaries; and the G-Mean ensures that majority-class performance is not sacrificed for minority-class gains. primary comparisons, in accordance with the limitations identified in prior literature [? ? ? ].

## 4 Experimental Results Summary

This section provides a comprehensive summary of the experimental results across all evaluated methods and datasets.

## 5 Detailed Experimental Analysis

In this section, we provide a detailed analysis of the experimental results, categorized by the level of intervention: data-level, algorithm-level, and hybrid strategies.

### 5.1 Data Level Methods

Data-level methods focus on reslicing the training set to achieve a more balanced distribution. This section presenting the G-Mean performance for individual classifiers and visual analysis of the results.

#### 5.1.1 *Experimental Tables: Data-Level Results.*

#### 5.1.2 *Visual Analysis of Data-Level Methods.*

This subcategory compares the G-mean performance across multiple classifiers (KNN, Random Forest, SVM) using different data-resampling techniques. Figures 1 through 4 illustrate the impact of oversampling, SMOTE, and SMOTE+ENN.

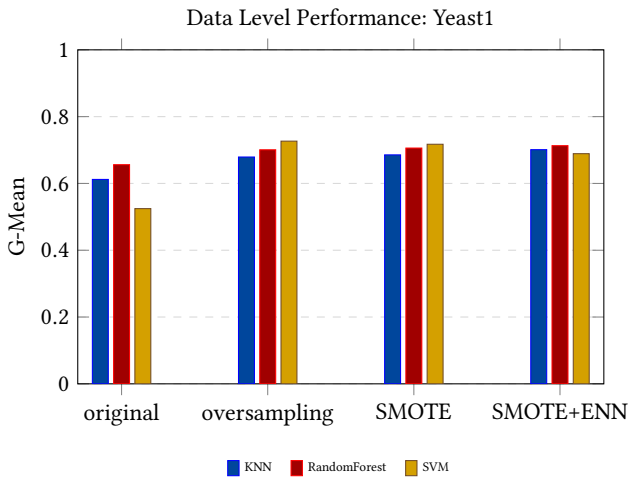


**Figure 1: [Data Level] G-Mean performance comparison on Yeast1 dataset.**

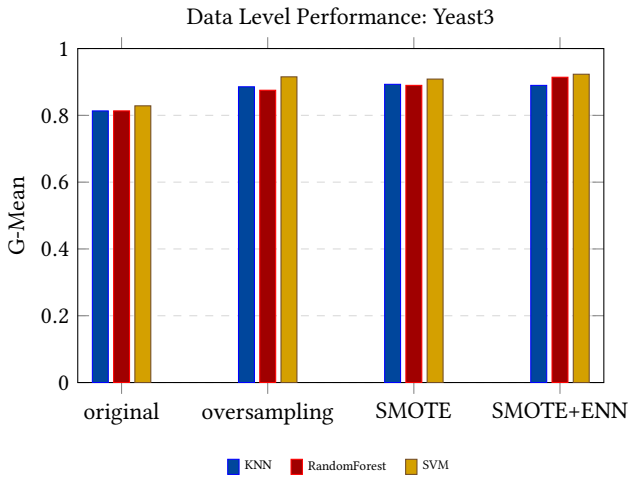


**Figure 2: [Data Level] G-Mean performance comparison on Yeast3 dataset.**

### 5.2 Algorithm Level Methods

Algorithm-level methods, such as Cost-Sensitive learning combined with ensemble classifiers (Random Forest, XGBoost, AdaBoost), show higher robustness across varying imbalance ratios by directly internalizing the cost of misclassifying minority instances.

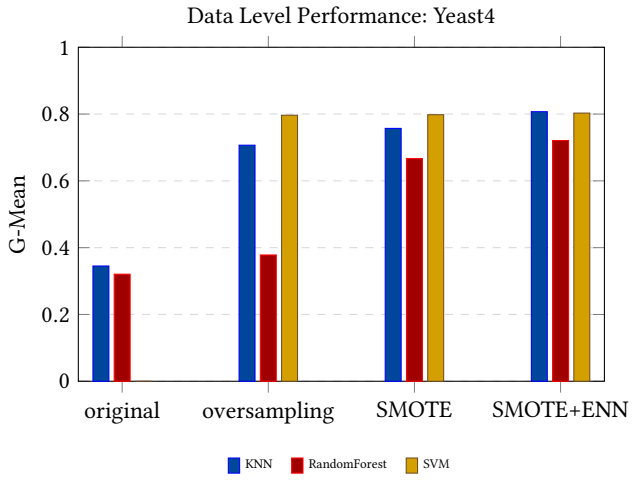


**Figure 3: [Data Level] G-Mean performance comparison on Yeast4 dataset.**

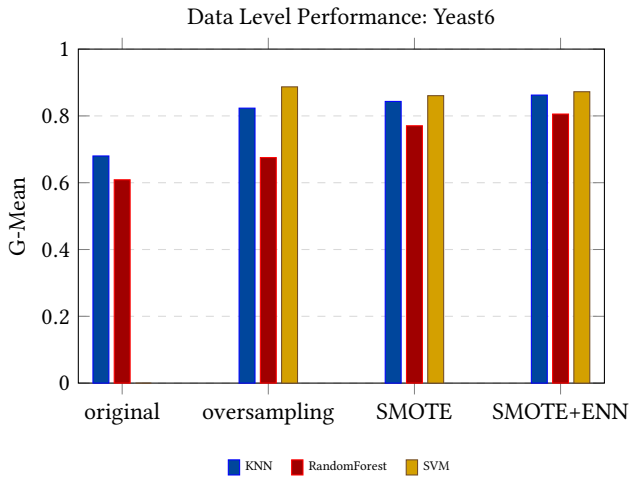


**Figure 4: [Data Level] G-Mean performance comparison on Yeast6 dataset.**

*5.2.1 Visual Analysis of Algorithm-Level Methods.* In this subcategory, we examine the performance shifts in ensemble methods when cost-sensitive mechanisms are applied.

Figure 5 presents a comparison of Recall scores between standard Random Forest and Weighted (Cost-Sensitive) Random Forest across the Yeast datasets.

Furthermore, Table 2 and Figure 6 highlight the "death cross" effect in AdaBoost. After applying cost-sensitive weights, Recall improves significantly, but at a severe cost to Precision.

Finally, the radar chart in Figure 7 illustrates the balanced improvement across multiple metrics (Recall, Precision, F1-Score, and G-Mean) for XGBoost when cost-sensitive learning is integrated.

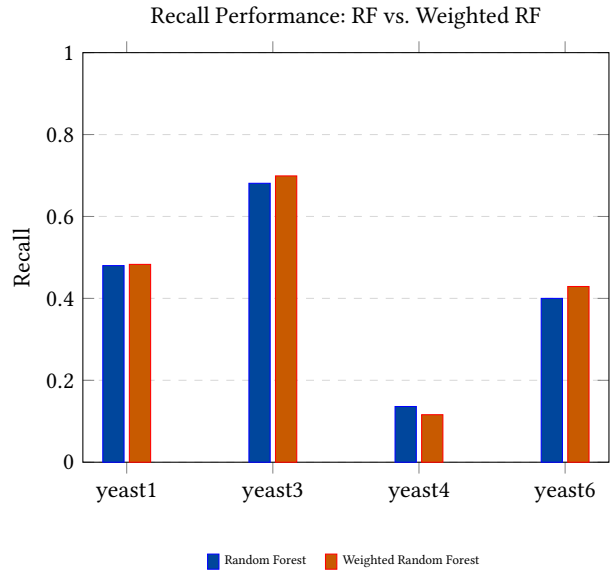


**Figure 5: [Algorithm Level] Recall comparison of standard and weighted Random Forest.**

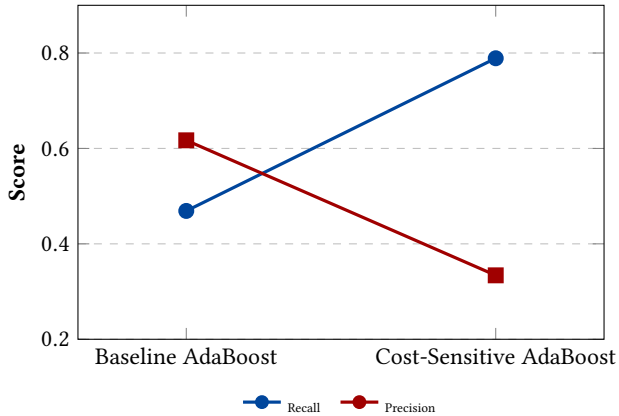


**Figure 6: [Algorithm Level] Dual-metric line chart illustrating the trade-off in AdaBoost.**

## 5.3 Hybrid Methods

Hybrid methods integrate data resampling with algorithmic enhancements. This subsection evaluates the efficacy of boosting-based hybrid approaches.

*5.3.1 Visual Analysis of Hybrid Methods.* The hybrid framework combines data-level resampling with algorithm-level boosting. Figure 8 and Figure 9 provide insight into these combined effects.

Figure 8 illustrates the F1-score performance of various hybrid boosting methods across four Yeast datasets.

*5.3.2 Impact of SMOTE Variants in Hybrid Frameworks.* Figure 9 shows the comparative performance of SMOTE-based hybrid strategies. SMOTEBoost demonstrates superior F1-scores compared to

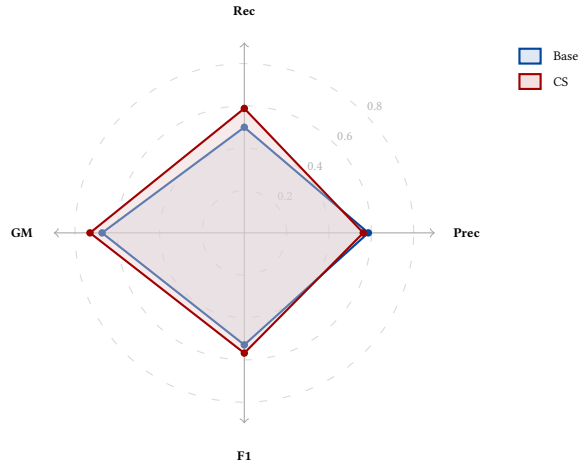


**Figure 7: [Algorithm Level] Radar Chart comparing Average XGBoost vs. CS-XGBoost.**

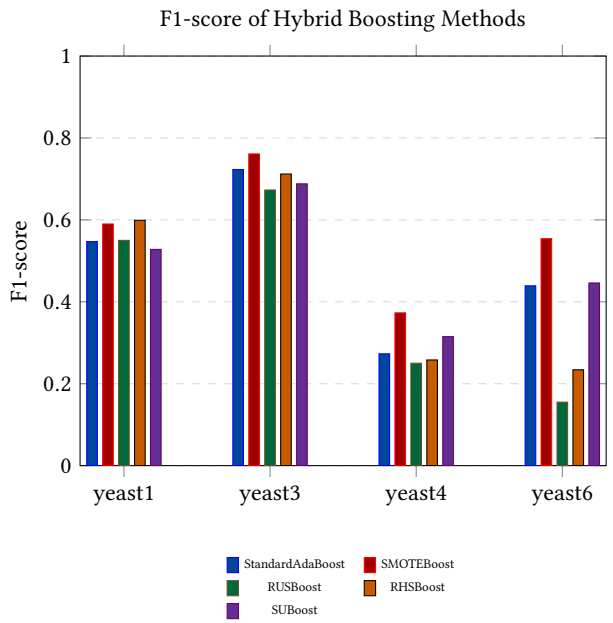


**Figure 8: [Hybrid Method] Performance comparison of various hybrid boosting methods (based on exp_1.csv).**

the pure data-level SMOTE and the cost-sensitive variant SMOTECSL, particularly as the imbalance ratio increases in Yeast-4 and Yeast-6.

*5.3.3 ROC Curve Analysis for Hybrid Frameworks.* Figure 10 presents the ROC curves of various hybrid boosting methods across the four Yeast datasets. The area under the curve (AUC) provides a summary metric for classification performance.

## 5.4 Summary Statement

The experimental results demonstrate that while data-level methods provide a foundational improvement, algorithm-level and hybrid

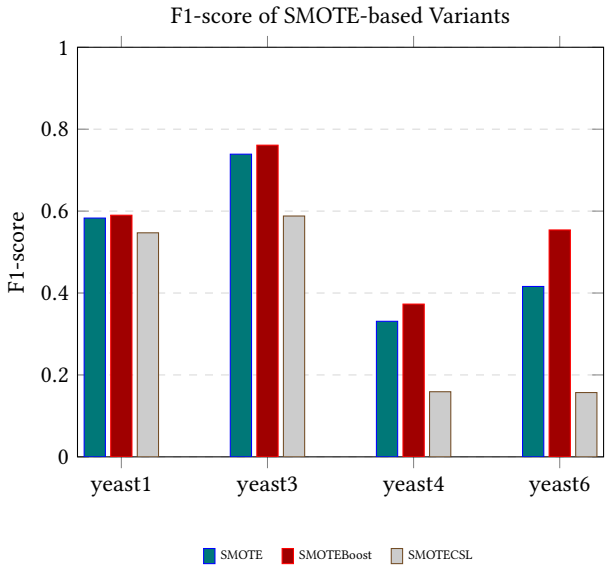


**Figure 9: [Hybrid Method] Impact of SMOTE variants on model performance (based on exp_2.csv).**

approaches offer superior stability and performance across extreme imbalance ratios.

## References

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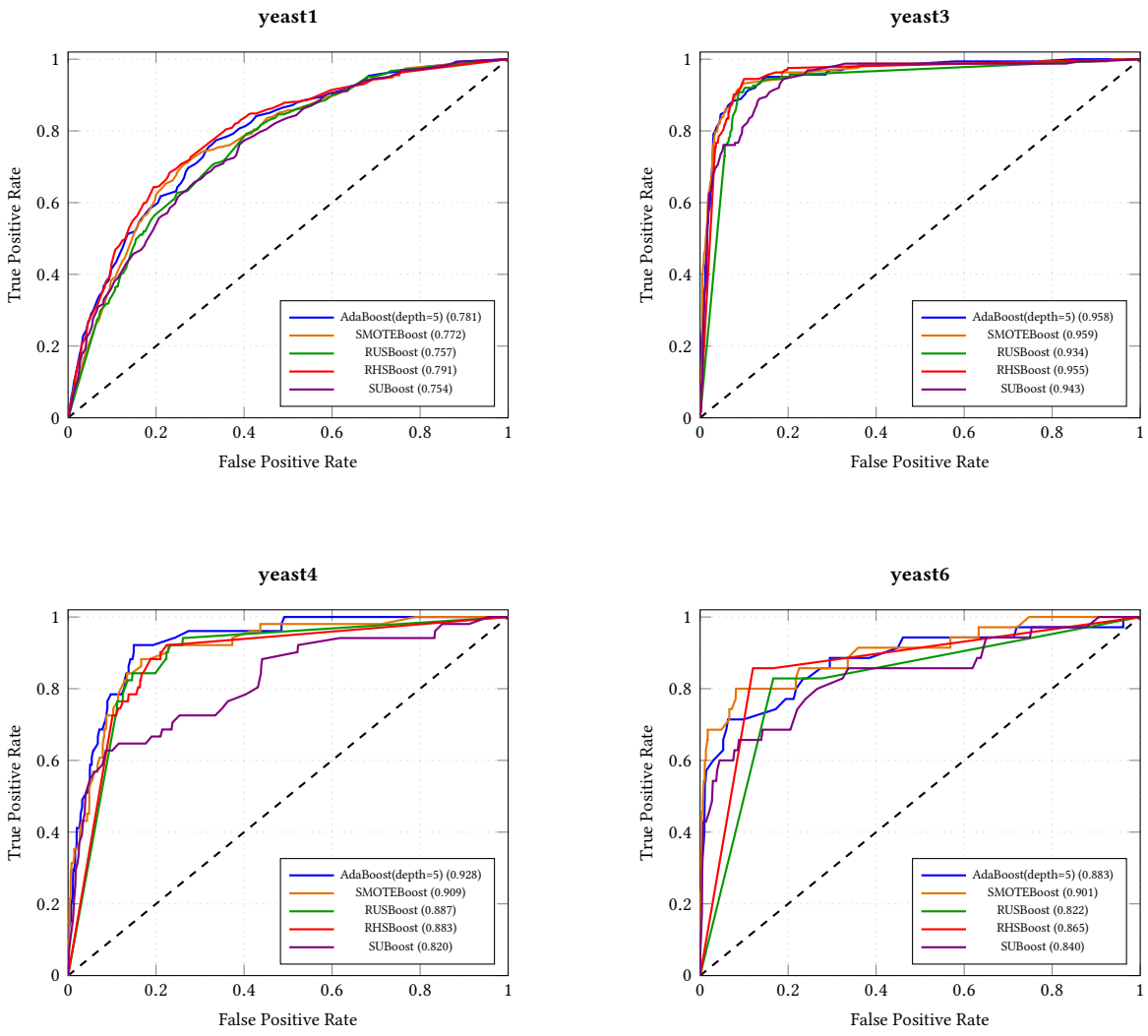


**Figure 10: [Hybrid Method] ROC Curve comparison across yeast1, yeast3, yeast4, and yeast6.**

**Table 2: Comprehensive Experimental Results Summary across all methodology levels.**

| Type | Method | Dataset | Acc. | Prec. | Rec. | F1 | AUC | PR-AUC | G-Mean | BAcc. |
|---|---|---|---|---|---|---|---|---|---|---|
| Data Level | Baseline | yeast1 | 0.7581 | 0.6410 | 0.4025 | 0.4850 | 0.7881 | 0.5887 | 0.5976 | 0.6526 |
| | | yeast3 | 0.9477 | 0.8106 | 0.6848 | 0.7412 | 0.9570 | 0.7924 | 0.8184 | 0.8325 |
| | | yeast4 | 0.9661 | 0.3556 | 0.0970 | 0.1403 | 0.8561 | 0.3739 | 0.2219 | 0.5470 |
| | | yeast6 | 0.9798 | 0.4556 | 0.2952 | 0.3445 | 0.8999 | 0.5656 | 0.4296 | 0.6458 |
| | oversampling | yeast1 | 0.7280 | 0.5356 | 0.6596 | 0.5841 | 0.7835 | 0.5737 | 0.7021 | 0.7077 |
| | | yeast3 | 0.9292 | 0.6479 | 0.8488 | 0.7290 | 0.9509 | 0.7530 | 0.8920 | 0.8940 |
| | | yeast4 | 0.9261 | 0.3308 | 0.4897 | 0.3022 | 0.8732 | 0.3626 | 0.6272 | 0.7157 |
| | | yeast6 | 0.9537 | 0.3796 | 0.6857 | 0.4355 | 0.9080 | 0.4895 | 0.7952 | 0.8230 |
| | SMOTE | yeast1 | 0.7201 | 0.5209 | 0.6760 | 0.5831 | 0.7834 | 0.5747 | 0.7029 | 0.7070 |
| | | yeast3 | 0.9328 | 0.6561 | 0.8551 | 0.7391 | 0.9571 | 0.7664 | 0.8971 | 0.8988 |
| | | yeast4 | 0.9115 | 0.2465 | 0.6152 | 0.3315 | 0.8801 | 0.3527 | 0.7406 | 0.7686 |
| | | yeast6 | 0.9452 | 0.3202 | 0.7333 | 0.4155 | 0.9240 | 0.4788 | 0.8248 | 0.8418 |
| | SMOTE-ENN | yeast1 | 0.6734 | 0.4657 | 0.7948 | 0.5846 | 0.7803 | 0.5666 | 0.7011 | 0.7094 |
| | | yeast3 | 0.9216 | 0.6028 | 0.8938 | 0.7177 | 0.9533 | 0.7438 | 0.9089 | 0.9094 |
| | | yeast4 | 0.8821 | 0.1900 | 0.6921 | 0.2928 | 0.8881 | 0.3371 | 0.7769 | 0.7905 |
| | | yeast6 | 0.9306 | 0.2587 | 0.7810 | 0.3724 | 0.9324 | 0.4875 | 0.8468 | 0.8576 |
| Algorithm Level | RandomForest | yeast1 | 0.7783 | 0.6650 | 0.4803 | 0.5553 | 0.8133 | 0.6435 | 0.6561 | 0.6899 |
| | | yeast3 | 0.9447 | 0.7870 | 0.6805 | 0.7275 | 0.9706 | 0.8316 | 0.8134 | 0.8289 |
| | | yeast4 | 0.9677 | 0.6000 | 0.1364 | 0.2062 | 0.9206 | 0.4073 | 0.3206 | 0.5668 |
| | | yeast6 | 0.9831 | 0.7667 | 0.4000 | 0.5054 | 0.8997 | 0.5629 | 0.6088 | 0.6986 |
| | CS-RandomForest | yeast1 | 0.7891 | 0.6979 | 0.4825 | 0.5688 | 0.8114 | 0.6420 | 0.6632 | 0.6981 |
| | | yeast3 | 0.9461 | 0.7896 | 0.6992 | 0.7396 | 0.9726 | 0.8479 | 0.8254 | 0.8379 |
| | | yeast4 | 0.9650 | 0.3333 | 0.1164 | 0.1655 | 0.9269 | 0.4298 | 0.2572 | 0.5557 |
| | | yeast6 | 0.9838 | 0.8667 | 0.4286 | 0.5305 | 0.8965 | 0.5907 | 0.6350 | 0.7129 |
| | XGBoost | yeast1 | 0.7648 | 0.6122 | 0.5151 | 0.5569 | 0.7847 | 0.6034 | 0.6662 | 0.6907 |
| | | yeast3 | 0.9434 | 0.7567 | 0.7176 | 0.7354 | 0.9612 | 0.7726 | 0.8338 | 0.8444 |
| | | yeast4 | 0.9629 | 0.4503 | 0.2764 | 0.3241 | 0.9165 | 0.3767 | 0.5024 | 0.6319 |
| | | yeast6 | 0.9771 | 0.5267 | 0.4857 | 0.4992 | 0.9246 | 0.5340 | 0.6875 | 0.7373 |
| | CS-XGBoost | yeast1 | 0.7561 | 0.5765 | 0.5989 | 0.5861 | 0.7859 | 0.6012 | 0.6998 | 0.7094 |
| | | yeast3 | 0.9468 | 0.7482 | 0.7852 | 0.7649 | 0.9615 | 0.7895 | 0.8709 | 0.8760 |
| | | yeast4 | 0.9582 | 0.4230 | 0.3945 | 0.3932 | 0.9107 | 0.4053 | 0.6066 | 0.6865 |
| | | yeast6 | 0.9764 | 0.4984 | 0.5714 | 0.5294 | 0.9242 | 0.5515 | 0.7399 | 0.7788 |
| | AdaBoost (d=1) | yeast1 | 0.7601 | 0.6293 | 0.4219 | 0.5045 | 0.7878 | 0.5760 | 0.6152 | 0.6598 |
| | | yeast3 | 0.9508 | 0.7891 | 0.7725 | 0.7770 | 0.9721 | 0.7808 | 0.8664 | 0.8726 |
| | | yeast4 | 0.9623 | 0.3500 | 0.1945 | 0.2439 | 0.9217 | 0.3758 | 0.3814 | 0.5920 |
| | | yeast6 | 0.9831 | 0.7000 | 0.4857 | 0.5655 | 0.9380 | 0.5549 | 0.6836 | 0.7404 |
| | CS-AdaBoost | yeast1 | 0.6658 | 0.4496 | 0.6922 | 0.5449 | 0.6736 | 0.4009 | 0.6729 | 0.6736 |
| | | yeast3 | 0.8956 | 0.5169 | 0.9203 | 0.6610 | 0.9470 | 0.7058 | 0.9061 | 0.9064 |
| | | yeast4 | 0.8922 | 0.2093 | 0.7436 | 0.3251 | 0.9190 | 0.2352 | 0.8154 | 0.8205 |
| | | yeast6 | 0.8956 | 0.1608 | 0.8000 | 0.2655 | 0.8972 | 0.2441 | 0.8374 | 0.8489 |
| Hybrid Method | AdaBoost (d=5) | yeast1 | 0.7540 | 0.5853 | 0.5196 | 0.5470 | 0.7821 | 0.5880 | 0.6612 | 0.6845 |
| | | yeast3 | 0.9421 | 0.7577 | 0.6928 | 0.7231 | 0.9591 | 0.7695 | 0.8199 | 0.8328 |
| | | yeast4 | 0.9650 | 0.5500 | 0.1964 | 0.2727 | 0.9312 | 0.4115 | 0.3923 | 0.5943 |
| | | yeast6 | 0.9778 | 0.5500 | 0.3714 | 0.4392 | 0.8979 | 0.4859 | 0.6043 | 0.6819 |
| | SMOTEBoost | yeast1 | 0.7473 | 0.5560 | 0.6294 | 0.5903 | 0.7718 | 0.5543 | 0.7074 | 0.7123 |
| | | yeast3 | 0.9468 | 0.7530 | 0.7725 | 0.7611 | 0.9596 | 0.7989 | 0.8641 | 0.8704 |
| | | yeast4 | 0.9535 | 0.3799 | 0.3946 | 0.3730 | 0.9147 | 0.4122 | 0.6148 | 0.6840 |
| | | yeast6 | 0.9778 | 0.5101 | 0.6286 | 0.5539 | 0.9147 | 0.5500 | 0.7704 | 0.8074 |
| | RUSBoost | yeast1 | 0.7156 | 0.5171 | 0.6085 | 0.5498 | 0.7558 | 0.5147 | 0.6728 | 0.6839 |
| | | yeast3 | 0.9003 | 0.5325 | 0.9203 | 0.6731 | 0.9348 | 0.5969 | 0.9090 | 0.9090 |
| | | yeast4 | 0.8255 | 0.1469 | 0.8418 | 0.2499 | 0.8847 | 0.1846 | 0.8321 | 0.8333 |
| | | yeast6 | 0.7864 | 0.0856 | 0.8286 | 0.1548 | 0.8265 | 0.0983 | 0.8009 | 0.8070 |
| | RHSBoost | yeast1 | 0.7507 | 0.5613 | 0.6456 | 0.5993 | 0.7910 | 0.5896 | 0.7148 | 0.7195 |
| | | yeast3 | 0.9191 | 0.5901 | 0.9015 | 0.7115 | 0.9546 | 0.6852 | 0.9111 | 0.9114 |
| | | yeast4 | 0.8396 | 0.1545 | 0.8036 | 0.2584 | 0.8824 | 0.1807 | 0.8210 | 0.8223 |
| | | yeast6 | 0.8679 | 0.1359 | 0.8571 | 0.2340 | 0.8675 | 0.1328 | 0.8580 | 0.8627 |
| | SUBoost | yeast1 | 0.7224 | 0.5297 | 0.5408 | 0.5277 | 0.7571 | 0.5586 | 0.6512 | 0.6685 |
| | | yeast3 | 0.9394 | 0.7743 | 0.6267 | 0.6877 | 0.9548 | 0.7286 | 0.7778 | 0.8024 |
| | | yeast4 | 0.9589 | 0.4417 | 0.2746 | 0.3146 | 0.8879 | 0.3961 | 0.4581 | 0.6289 |
| | | yeast6 | 0.9771 | 0.5343 | 0.4286 | 0.4462 | 0.8706 | 0.4566 | 0.6227 | 0.7095 |
| | SMOTECSL | yeast1 | 0.5492 | 0.3855 | 0.9416 | 0.5470 | 0.7931 | 0.5956 | 0.6053 | 0.6656 |
| | | yeast3 | 0.8538 | 0.4260 | 0.9510 | 0.5883 | 0.9688 | 0.8093 | 0.8947 | 0.8964 |
| | | yeast4 | 0.6874 | 0.0877 | 0.8618 | 0.1591 | 0.8535 | 0.1488 | 0.7648 | 0.7715 |
| | | yeast6 | 0.7540 | 0.0865 | 0.9429 | 0.1573 | 0.9100 | 0.3574 | 0.8363 | 0.8462 |

**Table 3: [Data Level] Summary of Experimental Results (Data-Level Methods)**

| Method | Dataset | Model | Acc. | Prec. | Rec. | F1 | AUC | PR-AUC | G-Mean | BAcc |
|---|---|---|---|---|---|---|---|---|---|---|
| Original | yeast1 | KNN | 0.7392 | 0.5623 | 0.4360 | 0.4904 | 0.7550 | 0.5083 | 0.6120 | 0.6492 |
| | | RF | 0.7783 | 0.6650 | 0.4800 | 0.5553 | 0.8133 | 0.6435 | 0.6561 | 0.6899 |
| | | SVM | 0.7568 | 0.6956 | 0.2910 | 0.4092 | 0.7961 | 0.6142 | 0.5246 | 0.6187 |
| | yeast3 | KNN | 0.9468 | 0.8119 | 0.6750 | 0.7362 | 0.9249 | 0.7188 | 0.8133 | 0.8277 |
| | | RF | 0.9447 | 0.7870 | 0.6800 | 0.7275 | 0.9706 | 0.8316 | 0.8134 | 0.8289 |
| | | SVM | 0.9515 | 0.8330 | 0.6990 | 0.7598 | 0.9755 | 0.8268 | 0.8285 | 0.8408 |
| | yeast4 | KNN | 0.9650 | 0.4667 | 0.1550 | 0.2147 | 0.7944 | 0.2928 | 0.3451 | 0.5741 |
| | | RF | 0.9677 | 0.6000 | 0.1360 | 0.2062 | 0.9206 | 0.4073 | 0.3206 | 0.5668 |
| | | SVM | 0.9656 | 0.0000 | 0.0000 | 0.0000 | 0.8532 | 0.4215 | 0.0000 | 0.5000 |
| | yeast6 | KNN | 0.9798 | 0.6000 | 0.4860 | 0.5281 | 0.8997 | 0.5167 | 0.6801 | 0.7387 |
| | | RF | 0.9831 | 0.7667 | 0.4000 | 0.5054 | 0.8997 | 0.5629 | 0.6088 | 0.6986 |
| | | SVM | 0.9764 | 0.0000 | 0.0000 | 0.0000 | 0.9003 | 0.6173 | 0.0000 | 0.5000 |
| Oversampling | yeast1 | KNN | 0.6894 | 0.4727 | 0.6573 | 0.5498 | 0.7382 | 0.4774 | 0.6791 | 0.6799 |
| | | RF | 0.7763 | 0.6244 | 0.5734 | 0.5964 | 0.8120 | 0.6383 | 0.7007 | 0.7161 |
| | | SVM | 0.7183 | 0.5098 | 0.7481 | 0.6061 | 0.8003 | 0.6054 | 0.7266 | 0.7271 |
| | yeast3 | KNN | 0.9131 | 0.5730 | 0.8525 | 0.6838 | 0.9151 | 0.6105 | 0.8855 | 0.8865 |
| | | RF | 0.9488 | 0.7556 | 0.7917 | 0.7729 | 0.9666 | 0.8332 | 0.8752 | 0.8799 |
| | | SVM | 0.9259 | 0.6151 | 0.9023 | 0.7303 | 0.9709 | 0.8153 | 0.9153 | 0.9156 |
| | yeast4 | KNN | 0.9360 | 0.2700 | 0.5491 | 0.3585 | 0.7934 | 0.2407 | 0.7068 | 0.7494 |
| | | RF | 0.9663 | 0.5444 | 0.1964 | 0.2624 | 0.9276 | 0.4124 | 0.3781 | 0.5950 |
| | | SVM | 0.8760 | 0.1779 | 0.7236 | 0.2856 | 0.8987 | 0.4346 | 0.7965 | 0.8025 |
| | yeast6 | KNN | 0.9589 | 0.3262 | 0.7143 | 0.4477 | 0.8954 | 0.3599 | 0.8233 | 0.8395 |
| | | RF | 0.9805 | 0.5993 | 0.4857 | 0.5177 | 0.8890 | 0.5812 | 0.6752 | 0.7391 |
| | | SVM | 0.9218 | 0.2134 | 0.8571 | 0.3411 | 0.9397 | 0.5274 | 0.8870 | 0.8903 |
| SMOTE | yeast1 | KNN | 0.6934 | 0.4783 | 0.6691 | 0.5575 | 0.7447 | 0.4872 | 0.6855 | 0.6862 |
| | | RF | 0.7642 | 0.5911 | 0.6013 | 0.5952 | 0.8043 | 0.6297 | 0.7059 | 0.7158 |
| | | SVM | 0.7028 | 0.4932 | 0.7575 | 0.5965 | 0.8011 | 0.6070 | 0.7173 | 0.7190 |
| | yeast3 | KNN | 0.9164 | 0.5817 | 0.8648 | 0.6944 | 0.9330 | 0.6695 | 0.8929 | 0.8938 |
| | | RF | 0.9481 | 0.7363 | 0.8223 | 0.7766 | 0.9666 | 0.8186 | 0.8898 | 0.8930 |
| | | SVM | 0.9340 | 0.6502 | 0.8780 | 0.7463 | 0.9715 | 0.8111 | 0.9086 | 0.9095 |
| | yeast4 | KNN | 0.9036 | 0.2039 | 0.6473 | 0.3077 | 0.8303 | 0.2104 | 0.7571 | 0.7800 |
| | | RF | 0.9522 | 0.3516 | 0.4745 | 0.3938 | 0.9107 | 0.3857 | 0.6667 | 0.7219 |
| | | SVM | 0.8787 | 0.1839 | 0.7236 | 0.2929 | 0.8993 | 0.4619 | 0.7980 | 0.8039 |
| | yeast6 | KNN | 0.9265 | 0.2144 | 0.7714 | 0.3344 | 0.9093 | 0.3032 | 0.8434 | 0.8509 |
| | | RF | 0.9778 | 0.5167 | 0.6286 | 0.5573 | 0.9183 | 0.5391 | 0.7705 | 0.8074 |
| | | SVM | 0.9313 | 0.2293 | 0.8000 | 0.3549 | 0.9444 | 0.5942 | 0.8606 | 0.8672 |
| SMOTE+ENN | yeast1 | KNN | 0.6671 | 0.4570 | 0.8042 | 0.5828 | 0.7481 | 0.4748 | 0.7011 | 0.7078 |
| | | RF | 0.7109 | 0.5007 | 0.7224 | 0.5903 | 0.7973 | 0.6146 | 0.7130 | 0.7143 |
| | | SVM | 0.6422 | 0.4392 | 0.8577 | 0.5807 | 0.7954 | 0.6105 | 0.6891 | 0.7061 |
| | yeast3 | KNN | 0.9003 | 0.5288 | 0.8771 | 0.6590 | 0.9222 | 0.6136 | 0.8897 | 0.8901 |
| | | RF | 0.9394 | 0.6726 | 0.8837 | 0.7628 | 0.9666 | 0.8096 | 0.9141 | 0.9150 |
| | | SVM | 0.9252 | 0.6070 | 0.9206 | 0.7312 | 0.9712 | 0.8083 | 0.9231 | 0.9232 |
| | yeast4 | KNN | 0.8760 | 0.1826 | 0.7455 | 0.2929 | 0.8424 | 0.1908 | 0.8071 | 0.8131 |
| | | RF | 0.9205 | 0.2305 | 0.5691 | 0.3253 | 0.9268 | 0.3933 | 0.7208 | 0.7511 |
| | | SVM | 0.8497 | 0.1570 | 0.7618 | 0.2600 | 0.8951 | 0.4273 | 0.8029 | 0.8073 |
| | yeast6 | KNN | 0.9036 | 0.1788 | 0.8286 | 0.2925 | 0.9077 | 0.2530 | 0.8624 | 0.8670 |
| | | RF | 0.9676 | 0.3899 | 0.6857 | 0.4941 | 0.9457 | 0.5976 | 0.8056 | 0.8301 |
| | | SVM | 0.9205 | 0.2073 | 0.8286 | 0.3305 | 0.9439 | 0.6119 | 0.8724 | 0.8756 |

**Table 7: [Hybrid Method] Summary of Experimental Results (Boosting-based Methods)**

| Method | Dataset | F1-score | ROC-AUC | G-Mean |
|---|---|---|---|---|
| AdaBoost(d=5) | yeast1 | 0.5470 | 0.7821 | 0.6612 |
| | yeast3 | 0.7230 | 0.9591 | 0.8199 |
| | yeast4 | 0.2727 | 0.9312 | 0.3923 |
| | yeast6 | 0.4392 | 0.8979 | 0.6043 |
| RHSBoost | yeast1 | 0.5993 | 0.7910 | 0.7148 |
| | yeast3 | 0.7115 | 0.9546 | 0.9111 |
| | yeast4 | 0.2584 | 0.8824 | 0.8210 |
| | yeast6 | 0.2340 | 0.8675 | 0.8580 |
| RUSBoost | yeast1 | 0.5498 | 0.7558 | 0.6728 |
| | yeast3 | 0.6731 | 0.9348 | 0.9090 |
| | yeast4 | 0.2499 | 0.8847 | 0.8321 |
| | yeast6 | 0.1548 | 0.8265 | 0.8009 |
| SMOTE | yeast1 | 0.5831 | 0.7834 | 0.7029 |
| | yeast3 | 0.7391 | 0.9571 | 0.8971 |
| | yeast4 | 0.3315 | 0.8801 | 0.7406 |
| | yeast6 | 0.4155 | 0.9240 | 0.8248 |
| SMOTEBoost | yeast1 | 0.5903 | 0.7718 | 0.7074 |
| | yeast3 | 0.7611 | 0.9596 | 0.8641 |
| | yeast4 | 0.3730 | 0.9147 | 0.6148 |
| | yeast6 | 0.5539 | 0.9147 | 0.7704 |
| SMOTECSL | yeast1 | 0.5470 | 0.7931 | 0.6053 |
| | yeast3 | 0.5883 | 0.9688 | 0.8946 |
| | yeast4 | 0.1591 | 0.8535 | 0.7648 |
| | yeast6 | 0.1573 | 0.9100 | 0.8363 |
| SUBoost | yeast1 | 0.5277 | 0.7571 | 0.6512 |
| | yeast3 | 0.6877 | 0.9548 | 0.7778 |
| | yeast4 | 0.3146 | 0.8879 | 0.4581 |
| | yeast6 | 0.4462 | 0.8706 | 0.6227 |